

Title: Framing Communities of Learning: Title Pages of Euclid's *Elements* and Early Modern Mathematical Education

Abstract: Euclid's *Elements* is a foundational work that shaped mathematical education for over two millennia. In early modern Europe, the treatise was translated and adapted dozens of times, serving as a core teaching text in universities and colleges, civic and military offices and professional training spheres. This paper examines how printed editions positioned the *Elements* within these settings by analyzing the title pages of 219 editions of the *Elements* published before 1700. Using a digital humanities approach, the study combines close reading with automatic and LLM-assisted classification of textual and visual features on title pages, enabling large-scale comparative analysis across languages, regions and institutional contexts. The title pages reveal how printers, editors and scholars framed and presented Euclid's *Elements* as a tool for instruction within varied communities of learning. The findings show the treatise as a canonical text in motion, animated by early modern scholarship through supplementation, abridgment, correction and debate. They also map the social terrain of its circulation, demonstrating strong local roots alongside active pan-European exchanges. In terms of intellectual priorities, adapters worked across several aims, balancing humanist ideals of philological recovery with strategies that sought brevity, clarity, accessibility, utility and innovation. By treating title pages as mediating spaces where intellectual, commercial and pedagogical concerns intersect, the study shows how paratextual and digital analysis illuminate the social and conceptual worlds in which early modern educational books took shape and complicate linear narratives of a unified Euclidean tradition, revealing instead a varied landscape of learning communities and practices.

Biography: Mia Joskowicz is a PhD candidate in the History Department at Tel Aviv University and a research engineer at Laboratoire SPHère (CNRS) in Paris. Her research examines early modern mathematical education and the printed transmission of Euclid's *Elements*, integrating traditional methodologies and digital humanities approaches. At Laboratoire SPHère she develops tools for large-scale comparative analysis of early modern editions of the *Elements*.